The mathematics first

The board is an infinite graph with states (x, h), where x is any integer and h is a nonnegative integer. A horizontal move changes x by $\pm 2^h$; an up or down move changes only h by 1. Every move costs one. All questions start at $(0, 0)$ and finish at height 0; the printed window is not a boundary.

For any ground-to-ground trip using at most N moves and reaching maximum height H, at least $2H$ moves are vertical. Its absolute displacement is at most

$$(N - 2H)2^H, \qquad 0 \le H \le \lfloor N/2 \rfloor.$$

This includes leftward motion, repeated climbs and horizontal moves on several levels. The farthest possible right endpoint for a budget N is exactly the maximum of this expression: climb to a chosen H, use all remaining moves to the right there, and descend. Thus reaching 16 needs **8 moves**. From a budget of 2 on, two more moves double the farthest reach (climb one level higher and every stride doubles): budgets $4, 6, 8, 10$ reach $4, 8, 16, 32$, so the shortest trip to 2^k is $2k$ moves for $k \ge 1$. This exponential gain is the distortion named below.

Children may discover that a longer-looking detour can save moves, and that "I found a short trip" differs from "no trip can be shorter." For destination 23, going to 24 and stepping left takes 10 moves, whereas every trip with no left moves needs at least 11. Overshooting 15 via 16 only ties the no-left optimum.

Who this fits. One Grades 4–5 prototype, requiring signed-coordinate tracking, adding and subtracting the printed step sizes, and a partner who can check a rule while recording one move word. Powers notation and general lower bounds are optional adult-supported reasoning. This nonnegative-height toy graph is related to exponential distortion in Baumslag–Solitar groups; it is not their full Cayley graph.

The other tables that hour. There is no younger packet and no tested age-fit claim; after the shared launch below, each table works from these pages. *Grades 2–3*, with an adult reading the rules: Problem 1, then Problem 2's question for target 8 on a spare board. Eight takes 6 moves, by exactly two trips, URRRRD and UURRDD, the same tie as 16's one level lower; it needs only adding 1, 2 and 4. *K–1*, adult-led on a spare page-1 board: the child moves the marker and counts dots while the adult says and tallies the moves. Is it faster to walk along the bottom to 4, or to go up, take big steps and come down? And to 6? Key in plain words: to 4 both take 4 moves (walk four, or up, two big steps, down); to 6, walking takes 6 but going up takes 5 (up, three big steps, down).

Materials and preparation

Use the older table of three children as a moving pair and rotating referee, with one anchored adult. Per pair: one four-page student packet, a spare board page, a stride strip cut from week-74-materials.pdf, pencil and eraser, and a very small pointed marker or pencil tip. Keep one move word as the recoverable record; count its letters after the trip.

Print the packet and the strips at actual size. Adjacent dots are 0.23 inches (5.84 mm) apart, so a level-4 stride spans 16 of them, too many to count by eye. The strip's ticks sit 1, 2, 4, 8 and 16 dot spaces from its 0. For R, put the 0 on the marker and move to the tick for the level's step; for L, put that tick on the marker and move to the 0. A normal broad pawn can obscure its position; check the chosen marker before the session. Rehearse U/D at an odd coordinate and R/L with the strip on an upper level, as in the launch. Vertical movement must preserve x. Use an extra sheet if a child goes beyond the window; do not forbid the move. No physical marker-fit, strip or partner-operation rehearsal has yet occurred.

Launch and a feasible first visit

Launch with all three tables together. Let children move the marker briefly. Show the student example beginning at coordinate 1, level 0: U reaches $(1, 1)$, R reaches $(3, 1)$, D reaches $(3, 0)$. The recorder writes URD, three moves. Then show one R from an odd coordinate on level 2 with the strip: at $(1, 2)$, lay the strip's 0 on the marker; the 4 tick lands on $(5, 2)$. Reset to $(0, 0)$ for the tasks. Have a child check a second legal attempt before

the pair works independently.

Allow about 10 minutes for Problem 1 and 15–20 for Problem 2, exchanging roles after a trip. Stop the first exploration with an eight-move route to 16 and an explicit distinction between "best found" and "proved best." Problem 3's budgets can supply a later proof investigation before Problem 4. Problem 5, especially the no-left lower bound for 23, is a readiness-dependent continuation or return visit; completing every page is not the session goal.

Answers and optional hints

Each word below begins at $(0, 0)$ and finishes on the ground. Other shortest words are welcome; replay them against the move rule rather than matching spelling.

Problem 1

Coordinate 3 takes 3 moves: RRR. Coordinate 7 takes 6: URRRDR. Two moves cannot reach 3: a ground trip stays at level zero unless its entire budget is spent going up and down, so its displacement is at most 2. Five moves reach at most 6, by the height bound with $H = 0, 1, 2$ giving $5, 6, 4$. Thus the proposed trips to 3 and 7 are optimal. *Hint:* ask what the elevator ride costs before counting the longer strides.

Problem 2

Two optimal eight-move words are UURRRRDD and UUURRDDD. They use four strides of 4 and two strides of 8 respectively. Problem 4 proves the lower bound, and these are the only two eight-move trips: equality in $(8 - 2H)2^H \geq 16$ forces $H = 2$ or 3, exactly $2H$ vertical moves, and every horizontal move rightward on the top level. A different claimed route of eight or fewer moves has a replay error; check it move by move.

Problem 3

Budget	Farthest coordinate	One attaining word
4	4	RRRR
5	6	URRRD
6	8	UURRDD
7	12	UURRRDD
8	16	UUURRDDD

For every permitted H, at most $N - 2H$ moves remain horizontal, each worth no more than 2^H. Maximize $(N - 2H)2^H$ over these finitely many heights. The ascent, right-strides, descent construction attains every corresponding bound, so the largest bound is the actual optimum. Using fewer than N moves does not evade the bound. *Hint:* "If the highest floor is H, which moves must already have been spent getting there and home?"

Problem 4

No trip of seven or fewer moves can reach 16 and return to ground. For seven moves, the complete height cases are

Maximum height H	0	1	2	3
Displacement bound $(7 - 2H)2^H$	7	10	12	8

All are below 16; reaching height 4 and returning already costs 8 moves. The triangle inequality is the key to the generality: the absolute value of a sum of signed strides is at most the sum of their absolute values. Repeated climbs use *more* vertical moves, intermediate floors give *smaller* strides, and left steps cannot exceed the absolute-value allowance. None makes a loophole. Together with Problem 2, this proves an exact optimum of 8.

Keep the argument concrete

Before writing powers, children can set aside two move tokens for every level climbed, then ask how far each remaining horizontal token could possibly go. These tokens represent an upper bound, not a requirement that all actual trips use only one level. Let the partner challenge a bound with a route and explain which allowance it uses. This preserves children's route choices while making the exhaustive height cases visible.

Problem 5

Target	Minimum	One shortest word	Minimum with no L
9	7	UURRDDR	7
15	9	UUURRDDDL	9
17	9	UUURRDDDR	9
23	10	UUURRRDDDL	11

Lower bounds for 9 and 17. Six moves reach at most 8, and eight moves reach at most 16 (Problem 3), so these targets need at least 7 and 9.

Why 15 needs 9. With budget 8, heights 0 and 1 allow only 8 and 12 displacement; height 4 spends all moves vertically. At height 2, at most four horizontal moves remain. An odd endpoint needs an odd stride, which is possible only on level 0, leaving displacement at most $1+4+4+4=13$. At height 3, only two horizontal moves remain; the analogous bound is $1+8=9$. These are all cases, so eight moves cannot reach 15.

Why 23 needs 10. For budget 9, the height bounds for $H=0,1,2,3,4$ are $9, 14, 20, 24, 16$. Only $H=3$ could reach 23. It permits at most three horizontal moves; one must be a ground stride to produce an odd endpoint. Their absolute displacement is at most $1+8+8=17$. So nine moves cannot suffice. The displayed ten-move word reaches 24 at height zero and then steps left to 23.

The no-left comparison needs its own proof. If the largest visited height is H, rightward strides are coins of values $1, 2, \ldots, 2^H$. The fewest coins totaling an integer target n is

$$c_H(n) = \lfloor n/2^H \rfloor + \mathrm{ones}(n \bmod 2^H),$$

where ones counts the 1s in a binary expansion. Combining two equal smaller coins into the next size shows that a minimum collection has at most one of each size below 2^H; the remaining smaller coins are precisely the binary remainder. Every trip therefore needs at least $2H + c_H(n)$ moves. This bound is attained by ascending to H and taking the required strides while descending through their levels.

For $n=23$, the bounds at heights $0,1,2,3,4$ are $23, 14, 11, 11, 12$. Heights at least 5 already use ten vertical moves and need at least one horizontal move, so cannot beat 11. The eleven-move no-left word UUURRDRDRDR gives $16+4+2+1=23$; the ten-move word with L is strictly better. For the other targets, no-left witnesses are UURRDDR for 9, UURRRDRDR for 15, and UUURRDDDR for 17. Since these attain the unrestricted minima, leftward motion cannot improve them. *Hint:* "Can passing the target by one help? Does it improve the best no-left route, or merely tie it?"

Source lineage and status

Nicholas Touikan, *An Introduction to Combinatorial and Geometric Group Theory*, Section 3.2.3, HNN extensions and dual tracks, discusses $BS(1,2) = \langle a, t \mid t^{-1}at = a^2 \rangle$ and its distorted cyclic subgroup. This supplies the exponential-shortcut connection. The stated graph, route instances and optimality proofs here are independently developed, not a literal Cayley-graph activity taken from that source. **Status:** matches GGT74-S-v2 and the GGT74-M-v1 strips (October 23, 2026). A search checks every answer and route; the proofs above cover all routes. Unpiloted; setup and pacing unrehearsed.